



LIFE SU-EATABLE

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Reducing carbon emissions in the EU through sustainable diets

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Executive Summary

The deliverables “*Guidelines for promoting sustainable diets at canteens of Worksites and Universities*”, “*Guidelines for promoting sustainable diets among customers in Food Retailers*” and “*Guidelines for promoting sustainable diets to the Citizen*” are the main reference documents resulting from the **preparatory action A1** “*Observatory on Sustainable Diets: analysis of the initiatives to promote a shift in dietary behaviour*”. The documents are the result of an extended analysis of the latest developments and past or on-going initiatives related to the promotion of sustainable diets among different focus groups, including civil society, governments and local areas, the private sector, institutions, universities and worksites. The massive amount of data analysed by the BCFN Observatory team span from scientific data related to the quantification of the environmental impacts of food items and dietary options to social studies of behaviour in relation to food and to documents and dissemination material describing on-going initiatives on sustainable diets among focus groups of interest for this project.

The guidelines are the basis for shaping the experiments to be conducted at the canteens in the course of the project, but can also be regarded as a basic “live document” which could be used to set the action for further replication activity by interested stakeholders during the course of the project and beyond. The “live” quality of the document is due to the continuous feedback and document implementation that will occur during the project as a result of “learning by doing”, as the experiments in the canteens and the experiences with stakeholders in the course of the project are meant precisely to define and identify the best strategies for communicating with people and engaging them toward adopting sustainable diets.

Part I of the guidelines is common to the three documents and describes and establishes the rationale for issues and topics analysed and proposed in Part II, which comprises the specific guidelines for each focus group.

“*Guidelines for promoting sustainable diets at canteens of Worksites and Universities*” is specifically meant to support the work of dissemination and engagement at canteens. As such, it can be used by private and public institutions, their canteen managers, as well as by the catering companies that interact with canteens.



SU-EATABLE DIET GUIDELINES PART I

Premise

The SU-EATABLE LIFE project aims at contributing to the international initiatives on sustainable diet adoption by all the actors involved in food distribution and consumption, from citizens to canteens, retailers and public bodies such as municipalities.

The core actions of SU-EATABLE LIFE are designed to engage EU citizens to adopt a sustainable and healthy diet, leading to a substantial reduction in GHG emissions and water saving, through a series of pilot experiments performed at universities and company canteens and by means of an easy-to-use information system.

The SU-EATABLE guidelines define the basic principles underlying a sustainable diet aimed at GHG reduction and water saving and provide indications to support replication activity tailored to the three main types of actors involved in the offer of food to the citizens: the catering sector (for public and private canteens), food retailers and municipalities. Key concepts are derived from the analysis of scientific literature, public case studies and interviews.

The guidelines are comprised of a more general introductory section ("Part I") followed by a specific section (Part II) for canteens, food retailers and municipalities.

"Part I" briefly introduces the reader to the link between dietary choices, environmental sustainability and health. It concisely analyses the most common examples of sustainable diets and states the definition of "sustainable diet" adopted in the SU-EATABLE LIFE project, which is summarized in 8 key principles. Each principle has been chosen on the basis of its environmental and health benefits, based on scientific evidence. Suggestions for citizen engagement are also briefly discussed.

In Part II, the 8 principles are considered in the perspective of the specific stakeholder analysed; past and on-going case studies are reported or analysed as examples of successful initiatives to inspire future replication activities.

SU-Eatable diet: taking care of our health and planetary sustainability

We human beings are unequivocally linked to our planet, as together with all the other organisms, we interact with the Earth in a self-regulating complex system, which has allowed life on the planet (Lovelock 2009). Human actions are causing unprecedented changes on the planet (Lovelock 2009), to the point that we define the present geological era as "anthropocene" (Crutzen & Stoermer 2000). We use the planet's resources to create our food and our commodities and to derive the energy we need for transport, heating, cooling and industrial processing. We consume land and alter natural ecosystems for urbanization,



agriculture, transportation, waste disposal, and industrial settlements, and for withdrawal of primary resources like metals, minerals, fossil fuels, wood, wildlife, and water. We release into the environment all types of pollutants, which redistribute in all the planetary compartments: air, land, water, biota.

In doing so, we are also shaping our future and the future of the coming generations. We are modifying the climate (IPCC 2018), we are impacting natural ecosystems with a drastic reduction of biodiversity (FAO 2019), and we are polluting and consuming critical resources like soil and water for food production at unprecedented rates (Rockström et al. 2009).

Each choice we make every day has a consequence on the planet and on ourselves

One of the most effective actions we can take as world citizens to support a more sustainable lifestyle is to act on our dietary choices (Willet et al. 2019). Food is an extremely important component of our daily life, in terms of both health and the impact we have on the planet. A change in dietary habits, as well as in the food distribution system, towards more healthy and sustainable options could be a relatively easy way to improve our lives and the health of the planet compared with more complex initiatives like shifts in transport mode or energy provision, where technological, political and infrastructural issues can have a much more prominent impact than single citizen choices.

Issues related to food, diet and sustainability cover a variety of aspects. Several of these aspects depend heavily on the food production sector, including the use of pesticides, GMOs, the farming system (organic vs. conventional), and the choice of quality standards for food production that protect the environment and reduce environmental pressures.

We, as citizens, can freely make dietary choices that can lead to significant environmental gains

Sustainable dietary choices made by millions and even billions of people can have striking effects on the amount of greenhouse gases we produce in the food chain systems and the amount of water we consume. By simply adopting more healthy and sustainable diets and paying attention to the way we get our food and beverages, we can take care of our health and significantly contribute to some of the most important objectives of sustainability of our times (Willet et al. 2019).

What do we know about the link between dietary choices and health? Or dietary choices and their environmental impact? Is there a clear link between what is good for our health and for the planet?

Dietary choices and health

A good diet is important for our health and well-being. It can also protect us against malnutrition in all its forms as well as non-communicable diseases. However, increased production of processed foods, rapid urbanization and changing lifestyles have led to a shift in dietary patterns. People are now consuming more foods high in energy, fats, free sugars and salt/sodium, and many people do not eat enough fruit, vegetables and other dietary fibre such as whole grains (WHO 2019). The growing prevalence of obesity is a global health concern, as it forecasts increasing incidence of several debilitating chronic diseases, including type 2 diabetes, hypertension, coronary heart disease, metabolic syndrome, respiratory conditions, cancer,



osteoarthritis, and reproductive, gall bladder, and liver diseases (Butland et al. 2007, Wang et al, 2011).

How do we define a healthy diet?

The exact make-up of a diversified, balanced and healthy diet will vary depending on individual characteristics (e.g. age, gender, lifestyle and degree of physical activity), cultural context, locally available foods and dietary customs. However, the basic principles of what constitutes a healthy diet remain the same based on evidence drawn from research, clinical expertise and reliable institutions.

According to the WHO, a healthy diet includes the following (WHO 2019):

- Fruit, vegetables, legumes, nuts and whole grains.
- At least 400 g (i.e. five portions) of fruit and vegetables per day.
- Less than 10% of total energy intake from free sugars.
- Less than 30% of total energy intake from fats. In particular, industrially produced trans fats are not part of a healthy diet and should be avoided.
- Less than 5 g of salt intake per day. Salt should be iodized.

Healthy diets are generally considered to include a diversity of nutrient-rich foods, such as vegetables, fruits, whole grains, and pulses (beans, legumes, nuts and seeds), with (for non-vegetarians or vegans) modest amounts of meat and dairy and unsaturated vegetable oils (Glopan 2016).

For the present guidelines, the indications for a healthy diet are based on recommendations by, among others, the World Health Organization (WHO 2019), the Joint WHO/FAO Expert Consultation on diet, nutrition and the prevention of chronic diseases (Geneva 2002), the British Dietetic Association (BDA), and the EAT-Lancet Commission on Food, Planet, Health. Considerations on health effects of specific food groups are based on proven evidence reported in the scientific literature.

Dietary choices and impacts on the environment

When we refer to our diet and the food we eat, we have to consider that each single food item is part of a full food supply chain (Poor & Nemecek, 2019; Willet et al. 2019). This chain starts from the farms where food items or their ingredients are produced, and includes the materials needed, the transport of materials to the farm and transport of products from the farm to processing stations, packaging, distribution to retailers and markets, and storage at retailers, often with refrigeration. The final step of the chain involves the management and/or disposal of used packaging and food “wasted” on the farms and at retailers.

In reality, rather than a food supply chain we can imagine that each food item is connected to a full network of interactions, each interaction having its own environmental weight in terms of resources and energy consumed, waste and pollutants released into the environment.

The most significant impacts of the food sector are related to the production of greenhouse gases (GHGs), which are key drivers of climate change (question box n°1), accounting for about 26% of all man-made GHG emissions (Poor & Nemecek 2019); water use by irrigation for agriculture, accounting for 70% of freshwater withdrawals; freshwater and groundwater pollution with chemicals, including excess P and N



(Willet et al. 2019); land use and biodiversity losses due to deforestation to convert natural land into croplands and pastures (Poor & Nemecek 2019, Willet et al. 2019).

Unsustainable rates of overfishing and excessive expansion of aquaculture in specific areas with intensive systems have also resulted in marine and freshwater degradation and biodiversity losses (Klinger & Naylor 2012, Willet et al. 2019).

Not all the food and food chains have the same impact on the environment

There is a growing amount of scientific evidence that shows that specific dietary choices might be helpful in reducing our impact on the planet.

The first important fact is that different food types have different levels of impact on natural resources (Clune et al., 2017, Poor & Nemecek 2019, Willet et al. 2019). Plant-based food is generally less impacting on climate and water resources than animal-derived food (Mekonnen & Hoekstra 2010, Clune et al., 2017, Willet et al. 2019). This can be explained by the fact that animal lifetime before slaughtering is on the order of several months to a few years. During these months the animals will need feed and care, so all the GHG emissions and water use associated with the cultivation of animal feed will be added into the calculation of the impact of the meat. Similarly, all the energy or water necessary to support the animal's feeding and well-being over the months will be accounted for on the estimate of the meat's environmental impacts.

Additionally, ruminants like bovines and sheep produce in the rumen a powerful greenhouse gas, methane (CH₄), which has a warming effect on the atmosphere 34 times higher than the gas CO₂ (IPCC, 2013). Hence beef and lamb meat have on average a greater impact on the climate than most other meat types (Mekonnen & Hoekstra 2010). Similarly, dairy products like cheese and butter that concentrate milk can also have a significant impact on climate and water resources (Mekonnen & Hoekstra 2010, Clune et al., 2017, Willet et al. 2019).

We can significantly reduce our footprint on the planet by balancing our diet, preferentially consuming food items with reduced environmental impact

A second important aspect related to the impact of the food sector regards farming practices. The main actors in this case are the food producers. The same food item can be produced using many different agricultural practices, which can result in differences of environmental impact up to 50-fold (Poor and Nemecek 2019). While this may be bad news on the one hand, on the other hand it clearly leaves wide space for the improvement of food production to meet environmental targets (Poor and Nemecek 2019, Willet et al. 2019).

Can consumers have an impact on this aspect of the food chain?

The answer is "YES". Consumers have the power to drive market transformation with their choices. Consumers should increase their awareness on issues like quality, traceability and sustainable production of food, and they should reward producers who value *quality above quantity*. Adoption of sustainability standards on a large scale could virtually force producers to adhere to such standards in order to gain market access (Poor and Nemecek, 2019).

A third aspect, as a more general perspective, is related to "the need for technological development in agriculture to achieve 'sustainable intensification'" (EU SCIENCE HUB), i.e. to produce more food per



hectare of land respecting sustainability standards. This technological development includes practices such as precision farming and smart agriculture, the use of agrobiodiversity for climate adaptation strategies, innovative breeding, crop improvement, and nanotechnologies or nanomaterials in farming for improved fertilizers and crop protection. The biggest challenge of this third aspect is the investment of public money to support technological development first and then adoption of the new technologies by farmers (Poor & Nemecek, 2019; Willet et al. 2019).

Citizens can drive public opinion by making their commitment to sustainable food choices clear at all levels of public engagement

The global call for a sustainable diet

Scientists from the health and environmental sectors have come to an agreement that the majority of the planet's population, and in particular in wealthy and developed countries, have average dietary habits that are both unhealthy and unsustainable for the environment (Willet et al. 2019). Most importantly, they have also underscored that this negative aspect leaves plenty of space for a significant change in dietary habits towards choices that would be beneficial to human health and contribute to keeping most of the critical environmental targets by 2050 within the so called "planet boundaries", i.e. "the global biophysical limits that humanity should operate within to ensure a stable and resilient Earth system—i.e. conditions that are necessary to foster prosperity for future generations" (EAT-Lancet Commission Summary Report: Healthy Diets From Sustainable Food Systems, 2019).

More recently, the commitment of governments and citizens on global environmental issues is reaching a new level of engagement, driven by the dramatic evidence of important environmental crises like global warming and climate change, the unprecedented extent of plastic pollution in the oceans, water scarcity, soil loss and degradation, and loss of productivity. In this context, a global call for sustainable diets is timely and responds to several of the seventeen Sustainable Development Goals (SDG), which the United Nations have set as key development strategies to be reached by 2030 in order to have a sustainable and equitable world (the 2030 AGENDA FOR SUSTAINABLE DEVELOPMENT). Sustainable diets could also contribute to meeting the strict requirements of the Paris Agreement to keep global warming to well below 2°C and achieve carbon emissions neutrality by 2050.

What is the current definition of a sustainable diet?

Many authors have tried to define the concept of "sustainable diet" with a scientific approach. The "reference individual" is an average adult male/female citizen with no specific pathologies. There are several environmental targets which are analysed in relation to dietary changes, the most common being the amount of GHGs emitted to produce one unit of food item (C footprint, question box n° 2), the water that has been used and polluted by all the processes involved to provide us with the food item (the water footprint, question box n°3), and the surface of land used by agriculture.

A summary of the most recent and relevant examples of what is considered a "sustainable diet" is shown in Table 1.

Table 1 – Most relevant examples of sustainable diets proposed by the scientific literature and/or related projects.

Authors	Diet specifics	Targeted environmental issues
Willet et al. 2019	Diversity of plant-based food; low amounts of animal source foods; use unsaturated rather than saturated fats; use limited amounts of refined grains, highly processed foods and added sugars; reduce food loss.	GHG, land use, water use, N&P use, biodiversity
Poore et al. 2018	Reduce meat consumption by half, in particular beef, lamb, farmed crustaceans; reduce cheese; lowering consumption of more discretionary products (oils, sugar, alcohol, and stimulants), use traceable food with land sustainability evidence.	GHG, eutrophication, acidification, water scarcity, land use
Garnett 2014	Food diversity; energy balance; whole grain; field grown vegetables; small quantity of meat; moderate use of dairy; very limited high fat food; limited sugar and salt; oils and fats with Omega 3:6 ratio; tap water.	GHG, land use
Livewell, WWF-UK	More vegetables; increase food variety; reduce food waste; moderate meat consumption; increase vegetable proteins; buy certified food; avoid food high in fat, salt, sugar; avoid soft drinks.	GHG, water use, land use

In all cases, the most impacting diets are considered those that have excessive consumption of meat, in particular red meat, as well as an excess of cheese, especially from sheep and cows. In diminishing order of impact are the pescatarian, flexitarian, vegetarian and vegan diets (Poor & Nemecek, 2019). The most recent example of sustainable diet proposed by the EAT-Lancet Commission, based on extensive literature on foods, dietary patterns and health outcomes, identifies a *planetary health plate* comprised by volume of approximately half a plate of vegetables and fruits, while the remaining part, considered in calories, consists primarily of whole grains, plant protein sources, unsaturated plant oils, and modest amounts of animal sources of protein, which however are optional.

The SU-EATABLE LIFE diet: definition and criteria

Based on the evidence provided by both the health and environmental sectors, the SU-EATABLE project has adopted the definition of sustainable diet as reported by the Food and Agriculture Organization 2010 (WHO 2018),

‘Sustainable diets are those diets with low environmental impacts which contribute to food and nutrition security and to healthy life for present and future generations. Sustainable diets are protective and respectful of biodiversity and ecosystems, culturally acceptable, accessible, economically fair and affordable; nutritionally adequate, safe and healthy, while optimizing natural and human resources’



We are aware that sustainability has many different aspects and dimensions ranging from environment to health, economics and social issues. To define a single set of recommendations for pilot studies for replication that encompass all these aspect would be extremely complex.

The SU-EATABLE project has thus chosen to tackle two of the most important environmental issues that have proved to have potential benefits from the adoption of a sustainable diet on a large scale, as discussed in the previous paragraphs. The key environmental targets addressed are represented by the reduction of the amount of GHGs emitted into the atmosphere and the reduction of water use and pollution.

We have organized our concept of sustainable diet into 8 key principles that express actions which any citizen or stakeholder could put in place on different scales to save CO₂ and water, based on dietary behaviour.

Each of the 8 principles is based on scientific evidence indicating that the suggested action is beneficial to the reduction of GHG emissions and water savings. However, in defining the 8 principles we have also considered actions that are less impacting yet still have important implications for other environmental targets and for social and economic issues, such as, for example, the issue of plastic bottled water or single-use packaging.

Therefore, we think that the most effective actions for climate and water should be combined with less effective ones because of the extra educational value these have for citizens, in order to develop an approach to sustainability in broader terms. The 8 key principles of the SU-EATABLE LIFE project are in line with the 10 criteria most recently produced by the United Nations on sustainable food systems and with the basic criteria reported by the EAT-Lancet Commission.

The 8 key principles of the SU-EATABLE DIET

From the analysis of scientific literature on the impact of food on health and environment, our project has distilled 8 key principles that constitute the basis for a healthy and sustainable diet. These principles can guide the actions of citizens and all the stakeholders involved in the food sector (retailers, caterers, restaurants, canteens, municipalities).

1. Champion plant-based food in your diet

Make vegetables, legumes, nuts and fruit the basis of our daily diet.

2. Enjoy meat in moderate amounts

Enjoy meat in moderate amounts, especially red meat and processed meats.

3. Enjoy dairy products in moderation

Consume dairy products, including milk, in moderation.

4. Avoid too much food

Avoid having too much food on your table and in your daily diet to avoid excess calories and food waste.

5. Celebrate variety

Vary your diet with the seasons and enjoy regional products and local varieties.

6. Fresh is best (for you and for the environment)

Favour fresh and naturally prepared food in your diet.

7. Drink tap water whenever possible and safe

Drink plenty of water, choosing tap water over bottled water whenever it is possible and safe.

8. Reduce single-use!

Reduce, reuse and recycle food packaging by bringing your own bags, cups and cutlery whenever possible. Choose food with minimum packaging and recycle and reuse materials wherever possible.

The scientific reasons for the choice of each principle are reported in detail in Appendix I. Explanations are given regarding the impact of each choice on Health and Environment. In a few cases, the action is neutral in terms of health yet still has a significant impact in environmental terms.

Table 2 below summarizes the 8 points in terms of the relative importance of the different actions on different sustainability targets. The indication is qualitative and based on expert judgment and should be used for comparative purposes only.

Table 2 – Impact strength on environmental and health targets as a direct consequence of engaging in the 8 suitable principles. A plus sign indicates a positive impact on the target as a consequence of engaging in that specific action. The / sign stands for a neutral effect.

Principles	Climate change	Water footprint	Land use	Biodiversity	Diffused pollution	Health
1. Champion plant-based food in your diet	++++	+++	++	+	+	+++
2. Eat meat in moderate amounts	+++++	++++	++++	+++	+++	+++
3. Enjoy dairy products in moderation	++	+++	+++	++	++	+
4. Avoid too much food	+++	+++	+++	+++	+++	+++
5. Celebrate variety	++	++	+	++++	+	++
6. Fresh is best (for you and for the environment)	++	+	/	+	+	+
7. Drink tap water whenever possible and safe	+	+	/	+++	+++	/
8. Reduce single-use!	+	+	/	+++	+++	/



For the specific purpose of the SU-EATABLE project, the environmental targets of reduction refer primarily to the impact on **climate (carbon footprint)** and **water resources (water footprint)**.

The SU-EATABLE project has set ideal boundaries or limits of CO₂ emissions and litres of water that should be associated with one meal in order to be sustainable.

For the climate scenario, a personal emission per meal has been established that allows European citizens to keep within the Paris agreement reduction of 40% of greenhouse gas emissions by productivity sector by 2030. These emissions are set in a range of 0.80 - 1.0 kg CO₂e per meal per day, assuming two main protein meals per day.

For the water resource, the limit is set in relation to personal water use per meal such as to remain close to the planetary boundary of water use, sufficient to support future sustainable development of humankind as indicated by the EAT-Lancet Commission for population growth projection to 2050. This water limit is 700-1000 litres of water footprint per meal per day, assuming two main protein meals per day.

Engaging people in sustainable behaviours

Making the environment a priority starts at the individual level (OECD 2013). Along with government policies, international agreements, corporate measures and technological innovations, behavioural choices that people take daily indeed play a fundamental role in enhancing sustainability. In recent years, motivating people to engage in sustainable behaviours as a mean to address environmental challenges has thus become a priority for national and local governments, not-for-profit environmental organizations and researchers, resulting in the application of a wide array of behaviour change techniques. But what is the best way to engage people on these issues?

Elements of successful engagement programs

According to the Centre for Climate and Energy Solutions (C2ES), successful engagement programs often incorporate four main elements, namely:

- **Education.** People must be aware of a problem and/or the opportunity for action before any engagement can occur. Providing accessible and credible information for the users will help to avoid dismissal or scepticism and to ensure understanding.
- **Meaningfulness.** As a second key step, people must find motivation to cope with new challenges. Failing to induce a sense of purpose blocks further progress towards change. Long-term change must align with people's life values in order to occur. Any activity that invites people to reflect on their beliefs and actions, share them and receive feedback will be particularly suitable for increasing meaningfulness.
- **Empowerment.** Empowerment is a process through which people gain greater control over

decisions and actions affecting their health. People can respond positively to a call to action if they have the capacity and possibility to engage. If there are obstacles to action, the program should seek to empower people by highlighting options and setting out a clear path forward.

- **Call to action.** A successful call to action is inviting, clear, and well-communicated.
- **Recognition.** People stay more engaged when they have a clear indication of the impact of their actions and are given positive reinforcement for changing their behaviour.

Engagement approaches and tools

Different barriers have been found as the main reason for the failure of programs dedicated to citizen engagement on environmentally sustainable behaviour. Engagement tools can be used accordingly to overcome such barriers (McKenzie-Mohr, D. 2013) (Table 3).

Table 3 – Engagement tool categories for tackling key barriers to successful engagement on sustainable dietary choices.

BARRIERS	TOOLS
Lack of knowledge	Communication Social modelling
Lack of motivation	Commitment Benefits
Forgetting to act	Prompts
Lack of social pressure	Feedback
Structural barriers	Environmental alterations

The first step to engage people on environmentally sustainable food choices is to raise their level of knowledge on the related issues, as a lack of awareness, information and knowledge has been recognized as one of the main barriers to participation in pro-environmental behaviours (Peschel et al. 2016).

Written information represents perhaps the most common type of intervention to promote behaviour change, enabling a considerable number of people to be reached with low effort and costs. It includes the following tools:

- **informational fliers and brochures** that advocate the behaviour and explain how/why to practice it
- **signs/posters** displayed in prominent areas that prompt the correct behaviour, describing



its benefits and importance

- **reminders (prompts)** in the form of fliers, stickers or other means such as text messages, containing a short message that tells people what they should do and often also why
- **internet** (websites and newsletters), social networking sites and digital apps

To be effective in triggering sustainable behaviour change, it is not sufficient to correctly inform people, as research in the field has shown that programs that are *strictly* information-based have little likelihood of substantively changing behaviour. Interventions **synergically combining different strategies** appear indeed to be particularly effective, since they enable acting simultaneously on different behavioural barriers (McKenzie-Mohr 2013). These might include the following activities.

Gaining a commitment from individuals that they will adopt a certain behaviour by signing a verbal or written pledge (ex. memorandum, group statement, billboard in the canteen, internal messaging platform). The action driver is that inconsistent behaviour is commonly perceived as a socially undesirable trait (Cialdini, 1988). Employee pledge boards placed in canteens with the words “I will...” written on them, so that people can write down things that they could individually do to help reduce food waste, have been used with success by the Rockefeller Foundation as a way to reduce food waste in the offices. Written, public and voluntary commitment has been demonstrated to be one of the most effective ways to foster sustainable behaviours.

Learning through observation of the behaviour of others, especially when it is relevant, easily understandable and permits the observer to achieve meaningful and positive outcomes. For example, a champion (individual or group) can be identified among people engaged in the project on the environmentally sustainable diet and her/his results can be publicized and explained as an example of good practice.

Giving benefits (e.g., rewards, refunds and unit pricing programs, gifts, prizes, lottery tickets, discount coupons, etc.) to people as a result of a positive or proactive behaviour. Incentives encourage behavioural change among those whose intrinsic motivation is low and helps to overcome monetary, time or effort-related personal costs. To avoid the negative effect of rewards, that is, the exclusive association of a positive behaviour with external rewards, a lottery scheme can be applied, i.e. not every instance of a behaviour is rewarded. An example is the Norwegian recycling system, where you can write your name and address on the milk cartons returned for recycling to participate in a lottery.

Making a sustainable behaviour more convenient and easier to perform by modifying the physical environment in which that behaviour occurs. Environmental alterations have been demonstrated to be particularly effective in changing habitual behaviours, since they disrupt the environmental cues that automatically trigger the habitual performance of the behaviour. Examples are “a meat-free day” in the weekly schedule of a canteen, an easily accessible area where “green dishes” or “most sustainable food types” are offered in canteens and food retailers.

Providing either individuals or groups with feedback regarding their behaviour, along with a comparison with a predefined standard so as to show the difference from the standard and motivating them to fill the gap. Feedback can provide the individual or group with information regarding:

- o the effects of his/her/their current behaviour compared to past behaviour (individuals or group)
- o the behaviour of other individuals/social groups, evoking social comparison mechanisms that

activate upward/downward comparisons (individual or group)

Receiving feedback on the performance of an entire group (a team, a campus, an office or a university canteen among several of the same firm or institution) instead of the performance of a single individual adds the feeling of collective efficacy to the feedback motivation.

Performance (feedback) and rewards can be communicated on site using screens, billboards, posters, leaflets, newsletters, websites, social networking sites and web apps.

Common thinking on sustainable diet issues – a starting point for planning engagement activities

The core psychological flow to induce engagement entails 1) awareness - inducing understanding of the impact of sustainable food choices on the environment and health, 2) threat and coping appraisal - offering learning strategies about alternatives, and 3) deciding to act - removal of barriers and offering of incentives to initiate action. To correctly plan an engagement campaign on sustainable food choices, it is important to start from the point of view of the engaged individuals, who might have different ideas, sensitivities, knowledge, misconceptions, or personal ideological or economic barriers in relation to these three key engagement fields.

Findings from two projects on sustainable diets, LiveWell for LIFE by WWF-UK and FCRN (Garnet et al.) 2015, both highly relevant for the SU-EATABLE LIFE objectives, indicate the following key barriers for limited engagement of citizens regarding a sustainable diet:

- **LACK OF AWARENESS:** Lack of solid knowledge on climate change issues, on the role of consumers in using food, on the C footprint concept. Confusion among environmental terms like organic, natural, and environmentally friendly.
- **HABITS AS A MAJOR BARRIER:** Habits are a major barrier for changing behaviour, and even with strong motivation it is difficult to change existing habits.
- **TOO EXPENSIVE:** some foods such as fruit and vegetables are considered more expensive than energy-dense foods. The issue might be geographically relevant.
- **TIME-CONSUMING:** Most people think a sustainable diet takes longer to think about, prepare and cook
- **CULTURAL BARRIERS:** strong social and cultural attachment to meat which associates meat consumption to masculinity and healthier/vegetarian diet to less masculinity (Canada).

Evidence collected from the SU-EATABLE LIFE survey

A survey was developed and submitted to anonymous volunteer participants with 13 questions (Appendix II) on the topic of sustainable diets in order to evaluate their degree of AWARENESS, THREAT AND COPING APPRAISAL, and WILLINGNESS TO ACT. The questionnaire was proposed online to citizens in the UK and Italy in Spring 2019 for three weeks and there was a total of 802 respondents. The majority were from 20-60 years old (60% of which between 20 and 40); 70% were female; 63% were employed and

17% students, and only a minor sample was composed of retired and unemployed persons. 72% of the respondents had an omnivorous diet and 18% a flexitarian diet. Vegans and vegetarians overall accounted for 7.5% of the respondents. The results of the questionnaire are summarized in Table 2 (**Appendix III**).

Most frequent answers to the thirteen questions

- **Q1** The majority of respondents associate the concept of “sustainable diet” with the concepts of: food good for the environment; with low GHG emissions and water consumption; good for biodiversity; locally grown possibly without chemicals; with recyclable or biodegradable packaging. A sustainable diet is not generally associated with low meat consumption or with vegetarian or vegan diets. Certified food is less of an issue, as well as free-range meat, wild fish or food sold by sustainable retailers or farmers markets.
- **Q2-Q3-Q4** The majority of respondents stated that they would not be able to distinguish which types of food have low carbon and water footprints, or low impact on biodiversity and the environment in general.
- **Q5** Only occasionally (44% of the answers) are buyers able to identify locally produced food or ingredients, and the remaining 46% of the respondents are equally balanced towards yes or no answers.
- **Q6** Around 70% of the respondents would be able to recognize if the food they are eating or buying is in season.
- **Q7** Regarding customer priorities when buying food, they are most sensitive to health impact and food quality (21% and 23%), seasonality (17%), environmental issues and cost (12% & 12%), traceability (7%), ease in cooking (5%).
- **Q8-Q9** Almost 80% of the respondents said they are very well informed or quite well informed on climate change issues and risks, while less informed on sustainable food production and sustainable agriculture.
- **Q10** 99% of the respondents think that citizens might reduce their personal impact on the environment through their food choices.
- **Q11** More than half (56%) of the respondents would be willing to pay more for food with a low impact on the environment; around 25% would adopt a more sustainable diet only if there were no cost increase or the increase were contained within 10%; the remaining 18% if the cost increase were contained within 30%.
- **Q12** The majority of respondents (93%) are interested in learning more about how their food choices influence the environment.
- **Q13** The answers related to how people would like to be informed on sustainable diet issues are quite diverse. The most frequent answer (22%) indicated “information displayed in a shop/canteen/restaurant/other”; 36% would like to be informed via websites, apps, social media (answers almost equally distributed), only 5% of the respondents would like to join a group, and



only 7% would like to receive newsletters.

TIPS BASED ON QUESTIONNAIRE RESULTS

- Although informed on the issue of climate change, people are generally not knowledgeable about the extent to which different food items differ in their impact on environmental targets. This lack of knowledge might limit their ability to make independent choices in canteens or retailers. This is therefore an important gap to be filled.
- People seem to be aware of the link between diet and climate or other environmental issues, but not of the specific reasons, including a general lack of awareness of the association between personal meat consumption and climate change or water consumption. Focused informational campaigns could help to increase knowledge and the engagement level. This also leads to the following point.
- Higher prices for sustainable food are not perceived by the majority of respondents as one of the most important barriers to sustainable food choices. However, only a minority of the respondents were in the categories which might be the most sensitive to cost issues (retired 6%, unemployed 3%). The cost issue needs to be analysed in depth in particular for the retail sector, where retired and unemployed people might be more frequent.
- Despite the stated awareness of climate change issues and their link with diet, the respondents still indicated the quality of the food and its effect on health as their first priority when buying food. In this respect, it is important to have informational campaigns that show that the most effective dietary choices are beneficial to both health and the environment.
- People are ready to learn more about a sustainable diet, and the preferential way to get information is to have initiatives in the place where food is consumed or bought.
- Overall evidence from answers Q1-Q5 indicates that people have an idea about the general links between health and environment, however they do not have a clear idea about how this translates into everyday choices. This is a typical outcome of surveys on health as well, and causes the so-called intention-behaviour gap: people know something but they do not know how to translate this into action and consequently they do not change their practices. This concept needs to be at the core of any specific action aimed at inducing long-term effective change in people's habits. Facilitating activities that engage people to share their own story and find out for themselves how to change will make a valid contribution to closing the intention-behaviour gap.

SU-EATABLE DIET GUIDELINES PART II “PROMOTING SUSTAINABLE DIETS AT WORKSITES AND UNIVERSITIES”

The role of canteens in promoting sustainable diets

Canteens, cafés and cafeterias at worksites, universities and schools are the best environments in which to begin the education of citizens on healthy and sustainable diets.

“Retailers and food services shape the immediate environment in which people buy food”¹

They offer the unique opportunity to maximize the number of people who can be reached with a few simple actions and represent the perfect environment to develop all the key steps for successful engagement of citizens on the topic of sustainable diets:

EDUCATION/RAISING AWARENESS: Canteens can provide an environment where educational messages and actions can be consistently and repeatedly delivered over long periods of time.

CALL TO ACTION/EMPOWERMENT: Canteens can provide practical examples of the type of foods that constitute a healthy and sustainable meal and examples of sustainable actions related to food management.

INCENTIVE/RECOGNITION: Through different engagement activities, from earning simple points that lead to meal bonuses to gamification activities involving the performance of individuals or entire groups, canteens can stimulate sustainable behaviour beyond simple awareness-raising.

Moreover, the majority of studies and experiences reporting a canteen’s shift towards higher sustainability show that engaging in sustainability brings financial benefits. In fact, redesigning the menu and the food offer towards higher vegetable consumption and meat reduction, reducing food loss and waste, decreasing the overall waste materials, repurposing excess food into innovative food offers, and involving staff in the sustainable mission, will greatly increase the public image of the brand and help to rationalize food provisioning costs and waste management costs.

Overall, the final result will be a win-win effect. On the one hand, large-scale citizen education on sustainable diets at canteens can have cascade effects on the civil society as a whole; on the other hand, due to the massive amount of food processed by canteens, they can be hotspots for the reduction of environmental impacts in the food sector.

Canteens and food services can orient consumers’ choices by providing virtuous examples of sustainable diets and food management. Relevant actions that can be carried out in canteens have been discussed by

¹ Quote from EAT-Lancet Commission Report



the EAT-Lancet Commission (Willet et al. 2019) and are mostly consistent with the 8 principles of the SU-EATABLE DIET reported in Part I. Key elements could be:

- To offer sustainable diet options in the menu (see SU-EATABLE DIET principles 1-3)
- To reduce the portions of food to meet healthy diet requirements (offer less) (SU-EATABLE DIET principle 4)
- To increase the range and seasonality of foods (SU-EATABLE DIET principles 5-6)
- To reducing packaging or to introduce innovative solutions for grab and go meals (SU-EATABLE DIET principles 7-8)
- To restrict the advertisement and marketing of unhealthy, unsustainable foods
- To support positive discrimination of healthy diets from sustainable food systems

Standards of public and private sector food procurement should be guided by the need to improve diets and access to vendors providing healthy and sustainable products¹

Where no regulation by local authorities exists to address the supply of meals in workplaces, schools, and venues, food providers should use contracts and procurement policies that promote healthy diets from sustainable food systems.

Different actors of the food chain should be encouraged to network in order to facilitate sustainable procurements. Links should be created among sustainable food producers and caterers or food distributors. An example of successful networking is the “Sustainable Restaurant Association” in the UK, a non-profit membership organization whose mission is to support food-service businesses to work towards sustainability. The SRA acts as a central hub to connect sustainable actors and build meaningful connections, removing a fundamental barrier for food providers, which is the supply of trustworthy and certified sustainable food.

Sustainable diets that taste good and are culturally appropriate should be promoted¹

Chefs at public and private canteens have the essential task to create healthy and sustainable food choices that are both delicious and affordable in order to achieve dietary change on a large scale. Chefs are increasingly recognizing their role in this cultural revolution, actively participating in creating new menus and championing public campaigns on healthy and sustainable food.

Examples include on-going initiatives like the Sustainable Restaurant Association, which involves thousands of chefs in all types of restaurants and food outlets who are working together to demonstrate that good food can be produced sustainably (www.thesra.org); the World Food Day promoted by FAO, involving hundreds of chefs from all over the world to promote healthy and sustainable diets; the New Nordic Kitchen Manifesto (<https://www.norden.org/en/information/new-nordic-food-manifesto>) signed by 12 Nordic chefs in 2004, which indicates the basic principles of the new Nordic cuisine.

Quote from EAT-Lancet Commission Report

These include high quality ingredients that respect climate, landscape, water, sustainability of fisheries and animal welfare, and recipes that are healthy, seasonal, simple and fresh and which highlight regional and



local food. Independent initiatives of single chefs who use their blog or website to promote healthy and sustainable diets, providing suggestions and recipes, are also increasing exponentially.

Canteens should substantially reduce their own impact

Together with processors and retailers, food providers and canteens could and should substantially reduce their environmental impacts (Poor and Nemecek, 2018). **THE MAJOR FIELD OF ACTION CONCERNS FOOD LOSS AND WASTE.** Strategies could be put in place by canteens to reduce their front and back waste with a **significant return in terms of image and money.** The Business Case for Reducing Food Loss and Waste study (Hanson and Mitchell 2017) showed that for every \$1 (or other relevant currency) invested in food loss and waste reduction, half of the surveyed company sites realized a \$14 or greater return. Clowes et al. (2018), analysing data over three years of pre-consumer waste from 86 catering sites in 6 countries, showed that the average benefit-cost ratio for food waste reduction was higher than 6:1, with 64 and 80 percent of the sites having recouped their initial investment in the first and second year, respectively. The results were independent of the geographic location of the businesses analysed.

A first important step for canteens towards the food waste reduction goal is **to get informed on the environmental impact of the food they provide**, in order to plan waste reduction where it matters most (Poor and Nemecek, 2018). For high-impacting food such as beef, losses and waste at catering facilities may represent a significant environmental burden and may be more relevant than how the meat is packaged or its provenance (relevant to transport impact).

Secondly, canteens could put in place a series of **winning strategies**. Clowes et al. (2018), in their study “THE BUSINESS CASE FOR REDUCING FOOD LOSS AND WASTE: CATERING”, provide indications for effective actions:

- **MEASURE FOOD WASTE:** Quantifying food waste by means of smart scales and measurement systems makes it possible to create food waste inventories that help managers to prioritize waste hotspots and monitor progress over time.
- **ENGAGE STAFF:** Staff engagement was found to be a key variable for successful action. This included defining and providing guidance on waste reduction, and informing daily staff via meetings, casual conversations, formal training and peer learning opportunities. Rewording approaches for having met objectives were among the most effective actions for long-term staff engagement. Reasons for failure included negative staff motivation (blaming), staff turnover, and difficult menus. Integrating food waste prevention and tactics to achieve it into standard staff training on operating procedures was considered one of the most effective measures.
- **START SMALL:** Small-scale pilot projects provide an opportunity to test food waste reduction programs and to evaluate winning options, barriers, difficulties and limits with a minimum financial loss. Gradually expanding the scope of the program allows adequate time for peer learning and can demonstrate its financial benefits over time.
- **REDUCE FOOD OVERPRODUCTION:** Key actions included: recording consistently under-consumed menu items and consequent reduction of daily servings; being more diligent about a meal's potential head count in the perspective of waste reduction, making it possible to better forecast canteen needs and reduce overproduction.
- **REPURPOSE EXCESS FOOD:** Having a plan to repurpose leftovers can allow the kitchens to generate revenue from wasted food items.

When newly engaging in food waste-reduction programs, canteens surveyed by Clowes et al. (2018)



invested the bulk of financial costs up front to conduct food waste inventories, to train staff on new schemes of food handling and storage procedures, and to modify the existing menus. The initial investment brought steady financial benefits over time, starting from reduced food costs (buying less food), increased revenue from new creative dishes accompanied by “informational campaigns”, and lower waste management costs. For more detailed information on waste reduction strategies, we suggest reading the full report by Clowes et al. 2018.

Actions that go beyond the specific objectives of the SU-Eatable project because they are related to food preparation rather than dietary choices, but which could still help canteens to reduce their environmental impacts, include:

COOK-CHILL VS COOK-WARM CHAINS: Studies indicate that the environmental impacts of the cook-warm chain (after being cooked, the food is transported to the point of use in insulated trucks) is lower than from the cook-chill system (blast chilling, refrigerated storage, refrigerated transportation and regeneration or reheating) (Coley et al., 2009, Espinoza-Orias et al., 2011, Fusi et al. 2016).

ENERGY SOURCE: Where national energy sources are heavily based on a mix of coal and petrol, gas rather than electric appliances significantly reduce impacts (Fusi et al. 2016).

COOKING MODE: Cooking in pasta cookers saves up to 60% of energy and 38% of water compared to range tops and therefore reduces the impacts associated with pasta preparation by 34-66% (Fusi et al. 2016). Electric cookers are overall the best option compared to electric range tops, with their impacts being on average 43% lower than induction tops and 57% lower than electric range tops (Fusi et al. 2016).

The 8 principles of the SU-EATABLE diet: practical tips for canteens

In Part I of these guidelines we introduced the 8 principles that define a sustainable diet from the perspective of the SU-Eatable project. An extended explanation for each claim is given in Appendix I. Here we provide tips for implementing each principle in the canteen.

1. Champion plant-based food in your diet

Make whole grains, vegetables, legumes, nuts and fruit the basis of your daily diet.

Health claim: Based on the recommendations of the World Health Organization and other authoritative bodies on a healthy diet, we recommend whole grains, legumes, nuts and at least 500 g (i.e. five portions) of fruit and vegetables per day. Emphasizing nutrient-dense, plant-based foods is associated with significantly better health outcomes.

Environmental claim: Plant-based food emits on average significantly less CO₂ equivalents (harmful greenhouse gases) compared to animal-based food. The water footprint of vegetables, grains and fruit is generally lower than that of animal-based foods, and is about 25 times lower when compared to beef.



Practical tips:

- Legumes provide a wide variety of nutrients, and include peas, chickpeas, beans and lentils.
- Increase vegetable and legume offer, as single dishes and as ingredients of main courses (pasta and legumes/vegetables, vegetable soups, rice with vegetables and legumes) or side dishes.
- Increase offering of fruit as dessert and fruit and nuts as a snack.
- Potatoes do not count as a vegetable but as a starchy food like bread.
- Increase the offering of whole grains such as wholegrain pasta and bread, brown rice, spelt, barley, buckwheat, sorghum and millet.

A portion according to Italian guidelines²:

- Fresh fruit: 150 g (a medium-sized fruit like an apple, banana, pear, orange and similar size fruit or 2-3 small fruits like apricots).
- 80 g of salad, or 200 g of cooked or raw vegetables (equivalent to 2-3 tomatoes, or 3-4 carrots, or 2-3 zucchini).
- 30 g of nuts.

A portion according to UK guidelines³:

- Any one of these: 1 apple, banana, pear, orange or other similar medium-size fruit, 3 heaped tablespoons of vegetables, a dessert bowl of salad, 30g of dried fruit or 150 ml (one glass) of fruit juice or smoothie (counts as a maximum of one portion a day).

These five portions should consist of different types of fruit and vegetables to include different combinations of fibre, vitamins, minerals and other nutrients.

2. Enjoy meat in moderate amounts

Enjoy meat in moderate amounts, especially red meat and processed meats.

Health claim: Processed meats (such as hot dogs, ham, sausages, corned beef, canned meat, and meat-based sauces, even if locally sourced) eaten daily may increase the risk of colorectal cancer and stomach cancer. A minimum intake of these types of food that guarantees complete safety cannot be defined but, in any case, it would be advisable not to exceed one serving/week. Similarly, red meat consumption (i.e. beef, veal, pork, and lamb) is associated with an increased risk of colorectal, pancreatic, and prostate cancers,

² LARN, 2014 Reference intake levels of nutrients and energy for the Italian population
http://www.sinu.it/public20141111_LARN_Porzioni.pdf

³ Eatwell Guide, 2016.

https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/742750/Eatwell_Guide_booklet_2018v4.pdf



cardiovascular disease, risk of stroke and type 2 diabetes. Such risks can be significantly diminished by substituting other protein sources for red meat, which should not exceed two servings/week.

Environmental claim: The impacts of the lowest-impact animal products typically exceed those of vegetable substitutes. At the highest end of CO₂-eq emissions is ruminant meat, i.e. beef and lamb, which can be almost 10 times higher than poultry, eggs, rabbit and most fish, and 4 to 5 times higher than pork. In terms of pressure on freshwater resources, meals that include beef have 3 times the water footprint as those which include other meat protein sources (pork, chicken) or legumes.

Practical tips:

- Reduce portions of red meat (i.e. instead of a main course, offer it as a sauce for pasta).
- Reduce frequency of red meat during the week.
- Substitute red meat with other sources of protein, preferably legumes. Other substitutes include fish, white meat (i.e. chicken, turkey, rabbit), eggs.
- When choosing fish, avoid the Big 5 (cod, haddock, tuna, salmon, prawns) and give preference to local varieties.

3. Enjoy dairy products in moderation

Consume dairy products, including milk, in moderation, giving preference to low fat products.

Health claim: High intake of dairy products has been widely promoted because of their high calcium content. However, the optimum calcium intake remains uncertain, while there is evidence that the adverse effects of fat of animal origin might outweigh the positive effect of calcium intake on the calcium balance. Animal-based fats like butter, lard or bacon fat put you at a higher risk of cardiovascular disease than unsaturated plant oils. Therefore, you should cook with plant oils such as olive or rapeseed as much as possible.

Environmental claim: Plant-based milks, except for rice milk, have a lower carbon footprint than dairy milk. Cheese can have a carbon footprint (CF) and water footprint (WF) up to 6 times higher than the original milk. When compared with other protein sources, cheese has twice the WF of soy-based proteins, and both the CF and WF of cheese can be higher than those of many meat sources, including chicken, pork and fish.

Practical tips:

- Substitute butter with extra virgin olive oil (preferentially) or non-tropical vegetable oils such as rapeseed or sunflower oil. Avoid tropical oils such as palm or avocado oil.
- Avoid cream in food preparation.
- Reduce the frequency of cheese offered during the week (one-two times).



- According to the Italian guidelines, a portion of fresh cheese such as cottage cheese or ricotta is 100 g; a portion of aged (soft and hard) cheese such as cheddar or gouda is 50 g.
- Non-dairy sources of calcium include green vegetables high in calcium (e.g. broccoli, kale), legumes, and water.

4. Avoid too much food

Avoid too much food on your table and in your daily diet, to avoid excess calories and food waste.

Health claim: Energy intake should be in balance with energy expenditure. Excess energy intake is the main dietary cause of weight gain.

Environmental claim: Food waste accounts for about 20% of food produced in the EU, equal to 88 million tons a year, with an economic cost of 143 billion euros. Each kilogram of food brings with it the environmental impact generated during its production, processing, transport and packaging.

Practical tips:

- Reduce serving sizes to fit your optimal caloric intake.
- Place vegetables and legumes at the beginning of the serving flow.
- Choose minimally processed, whole grain products such as bread, rice and pasta. Fibre-rich foods increase satiety.
- Limit as much as possible foods that are high in fat and/or sugar, such as pies, gravy, cakes, pastries, biscuits, soft drinks, ice cream.

5. Celebrate variety

Vary your diet with the seasons and enjoy regional products and local varieties.

Health claim: Dietary diversity has been recognized as a key element of high-quality diets. Increasing the variety of foods across and within food groups is recommended in most dietary guidelines internationally, as a means to ensure adequate intake of essential nutrients and to promote good health. Preference for seasonal, fresh and minimally processed foods can in most cases maximize the content of protective nutrients and substances in the diet.

Environmental claim: Agrobiodiversity is an important strategy for facing the challenges of climate change, supporting food security, resource saving, and helping to bolster local economies.

Practical tips:

- Explore local varieties of field-grown fruits and vegetables.
- Rediscover traditional dishes and introduce them into the weekly menu.



- Respect seasonality for fruits, vegetable and fish.
- When choosing fish, avoid the Big 5 (cod, haddock, tuna, salmon, prawns) and check the Good Fish Guide to see if fish local to you is good to eat.
- To reduce salt intake, use herbs (basil, parsley, thyme, rosemary, sage) and spices (chili, curry, cumin) to enhance taste. Salt intake should be limited to less than a teaspoon of salt a day.

6. Fresh is best (for you and for the environment)

Favour fresh and naturally prepared food in your diet.

Health claim: The increasing proportion of ultra-processed foods in diets has been identified as a driver of excess energy intake.

Environmental claim: Food processing can add further pressure on the environment, due to the higher amount of energy and materials required, which leads to higher CO₂-eq emissions and resource consumption per kilogram of product.

Practical tips:

- Avoid foods high in fat, sugar and salt (i.e. sugar sweetened beverages and ready to go meals).
- Give preference to healthy cooking methods such as steaming, boiling, baking and poaching, and drastically limit the offer of fried foods.

7. Drink tap water whenever possible and safe

Drink plenty of water, choosing tap over bottled water whenever it is possible and safe

Health claim: Good hydration is vital for good health and well-being. On average we should drink 1.5-2 litres of water per day. Water safety and quality are fundamental to human development and well-being. The quality of bottled water is not necessarily superior when compared with municipal tap water.

Environmental claim: When safe tap water is available, the use of water in plastic bottles places an unnecessary burden on the planet. Bottling 1.5 L of water requires around 1.9 L of extra water for bottling operations, processes, packaging and transport. Plastic production relies heavily on oil, and in the EU the amount of plastic recycled in many countries is still less than 50% of the production.

Practical tips:

- Introduce water jugs and ensure access to bottle refill taps.
- Recycle the bottles after use.
- Encourage the use of environmentally friendly reusable water bottles.



8. Reduce single-use!

Reduce, reuse and recycle food packaging by bringing your own bags, cups and cutlery whenever possible. Choose food with minimum packaging and recycle and reuse materials wherever possible.

Environmental claim: Food packaging represents 25% of the plastic materials that leak into the environment. The estimated annual input of mismanaged plastic waste to the ocean is about 13 million tons, causing irreversible harm to natural systems. If opting for compostable packaging, ensure that facilities exist locally to adequately compost it and that canteen guests understand which is the correct bin to place their waste in.

Practical tips:

- Reduce food packaging in canteens.
- Reuse the materials whenever possible.
- Recycle all plastic after use.

The SU-EATABLE PLATE characteristics

Which types of foods or meals can I label with a green mark of SU-EATABLE sustainability?

The ideal sustainable diet should respect the 8 principles. Moreover, each meal should be balanced in terms of quantities and ingredients to remain within the impact boundaries reported in Part I of these guidelines, i.e. the total amount of food served in the main meals of the day should account for no more than:

Carbon footprint: 0.80 - 1.0 kg CO₂-eq per meal

Water footprint: 700-1000 litres per meal

These quantities can be considered as average values, meaning that through the weekly dietary plan some balance is allowed between days and dishes. This means that if, on day 1, the food choice exceeds the limit for whatever reason, on day 2 you can plan your menu so as to consume food items with a lower impact.

Since in most cases canteens and chefs are not in a position to determine the impact of each food item, here we suggest a series of food items that are eligible for a green flag. With a green flag we mean a dish or meal that respects the 8 principles and is approximately within the CO₂ and water use boundaries of sustainability identified and reported above.

1. All plant-based recipes have a green flag. Further environmental improvement can be achieved by using fresh vegetables and fruit and products not grown in greenhouses, especially in countries where



greenhouses need to be heated.

2. Legumes have a green flag. Legumes include peas, chickpeas, beans, lentils and more.
3. Cereal grains and their products have a green flag, but for nutritional reasons whole grains are to be preferred to refined grains, and among grains rice has the highest impact.
4. Fish dishes have a green flag, excluding the most impactful ones like cod, haddock, tuna, salmon, and prawns, which should be consumed in moderate amounts during the week or as a portion (for example, as part of a recipe). Choose to serve a wide variety of fish, but first check the Good Fish Guide to determine whether they are a 'good' fish to serve. Fish from intensive aquaculture should be avoided.
5. Poultry meat has a green flag as long as the portion remains within the advised daily consumption of around 100-150g.
6. Cheese-based recipes have a green flag as long as the cheese is treated as an ingredient, so its weight is generally around or below 50g (for example, as cheese cubes in salads, or as an addition to a baked potato, etc.). Combining cheese and meat in the same dish totals a significant environmental impact, so in most cases the recipe cannot be flagged green.
7. Egg-based dishes can be flagged green as long as the portion is a maximum of 2 eggs.
8. Beef, lamb and pork based recipes can be flagged green only if the dish is plant-led, i.e. the amount of meat within the recipe is minimal, around 15 g per portion.
9. Fruit salad has a green flag, with preference given to seasonal and local fruit. Cut fruit served in single-use plastic containers should be avoided whenever possible.
10. Sweets and desserts cannot have a green flag, mostly because they do not align with the recommendations for a healthy diet rather than in relation to environmental sustainability.
11. No sugar-sweetened or artificially sweetened beverage, or beverage in a plastic container, can be flagged green.
12. For all green-flagged dishes, the caterer should pay attention to provide the balanced quantity of food without excess, which could lead to food waste and extra calorie intake.

Some examples of the impact of different food choices

For those who would like to have an idea of the impact of different combinations of food, we have calculated the carbon footprint of an average meal that could be consumed in a canteen or at home. The portion of different food types is calculated on the basis of average recommended proportions established by international health guidelines as reported in Table 4.

Table 4 – Average meal composition by food weight and food type of different food items as suggested for an average individual on the basis of dietary recommendations (see Part I, section on dietary health choices).

Grains (pasta, bread, rice, other cereals)	80 g
Vegetables (any fresh or cooked vegetable)	200 g
Protein equivalents (meat, fish, eggs, poultry, cheese)	100-150 g
Fats (olive oil, seed oil, margarine, butter)	15 g
Fruit (fresh fruit)	150 g

As the biggest difference in meal impact comes from the choice of protein source, we have calculated the total carbon footprint of the meal, varying only the protein source, while maintaining all the other food ingredients fixed. The carbon footprint results are shown in Table 5.

Table 5 – Value of carbon footprint of meals having the composition of Table 2. In each column the protein source is different. In the rows, the carbon footprint value is cumulative for consecutive days from 1 to 5 (Monday to Friday). In the last column (**RB recommended boundary**) we have reported the cumulative value that an ideal sustainable meal should have (1 meal, max 1 kg of CO₂e).

kg CO ₂ equivalents/meal									
Lunch at canteen	beef	cheese	pork	fish	chicken	eggs	legumes	veg	1 kg CO ₂ /meal
Days	A	B	C	D	E	F	G	H	RB
1	2.9	1.3	0.8	0.8	0.7	0.6	0.6	0.5	1
2	5.8	2.7	1.7	1.6	1.3	1.2	1.2	1.0	2
3	8.7	4.0	2.5	2.3	2.0	1.8	1.7	1.4	3
4	11.7	5.3	3.4	3.1	2.7	2.4	2.3	1.9	4
5	14.6	6.6	4.2	3.9	3.3	2.9	2.9	2.4	5

What can we see from Table 5?

- In each row, you can see how the choice of different protein sources results in quite different impacts. You can immediately see how much more CO₂ is emitted when choosing beef in your meal rather than eggs or fish or legumes (ex. beans) as the protein source.



- You can compare the value of the meals with the sustainability limit value of kg of CO₂ (**RB recommended boundary**) reported in the last column. You can see that except for beef and cheese, all the other protein sources are close to or lower than the sustainability limit **RB**; however, this is only true if your meal is balanced in terms of quantities, i.e. if you respect the proportion of food reported in Table 4. If you eat a double amount of food, then only legumes and poultry come close to the **RB** limit. Hence, exceeding the amount of food intake increases the burden for both health and the environment.
- If you eat beef in each meal, at least once per day, from Monday to Friday, you can see at day 5 how far you are from the sustainability limit **RB** compared to other protein sources. Eating beef with high frequency is hence environmentally unsustainable.
- Table 5 helps you to better understand the suggestions we have given regarding which meal would have a green flag of sustainability.
- From Table 5 you can also see that you can act on your diet on a weekly basis to balance your impacts. If, for example, you eat beef on day one (column A), you are above the **RB** value (1kg CO₂) by 1.92 kg CO₂ (2.92-1). To rebalance your week, you can thus choose on the other days some meals that allow you to save your extra 1.92 kg CO₂. Choosing legumes allows you to save 0.42 kg CO₂ with respect to the recommended value (1-0.58). This means that if you choose legume-based meals for the other 4 meals, your final impact over 5 days will be “2.92 kg CO₂ x 1 beef meal + 0.58 kg CO₂ x 4 legume meals = 5.2 kg CO₂”, just slightly above the 5 kg recommended for 5 days.

Reasoning in this way, you can understand how to balance your meals not only for 5 lunches at the canteen but also for the whole week, over lunch and dinner, wherever you are.

Of course many other elements contribute to your weekly diet, like beverages, sweets, snacks, and breakfast. However, the table gives you an idea of how much you can affect your sustainability already by choosing a balanced source of protein and following the 8 SU-EATABLE points.

Concerning the two objectives of the SU-EATABLE project, i.e. the abatement of greenhouse gas emissions (climate impact) and the reduction of the water footprint (resource impact) of the food served at canteens, we could classify **principles 1 to 3 as the most effective**.

Creating the conditions for customers to choose and enjoy plant-based and to a lesser extent plant-led dishes during the whole week, possibly with multiple choices to satisfy their curiosity and need for variety, would make it possible to achieve important reduction targets. Any strategy that drives attention and stimulates the customer towards a green flag choice will make it easier to reach the reduction targets.

Green flag options should be sufficiently original and tasty to win against the habit of choosing the most popular meat-based dishes (mainly beef and lamb) on a routine dietary basis. Thematic days to discover new green ingredients offered in new original recipes can be fun and appealing. Providing leaflets, panels, and informational materials of any kind to accompany the green food discovery with a clear



understanding of “the Why” green is good for health and the planet might further boost the stimulating effect of new dishes and recipes.

In many cuisines, rediscovering traditional dishes, plant-based and plant-led, might help the consumer to take home and replicate the experience.

Cheese should be offered in small portions accompanying other vegetarian ingredients. Fish, when sustainably fished, is a better alternative to meat, even in its classical UK version of fish and chips.

The **second most effective SU-EATABLE LIFE principle is number 4** “avoid too much food”, namely reducing any form of overfeeding or food waste. Different strategies can be put in place to contribute to point 4, as evidenced in the previous paragraph. In canteens where food waste is significant (1/3 of the food offered), rationalizing the food offer and optimizing food recycling and waste management can be an extremely important asset to be combined with points 1-4 to meet sustainability targets.

Points 5-6, i.e. preference for seasonal and local food, could bring additional advantage to meet the environmental targets, although to a lesser extent. Fresh food is of major importance when refrigeration can be minimized or avoided. The overall effect, however, is lower than what can be obtained by acting on points 1-4. On the other hand, fresh and seasonal is often linked to the quality of the food items, in particular if the food is also organic, which helps the chefs to offer tasty food that can satisfy customers and win the comparison with meat dishes.

Points 7 and 8 mostly concern the reduction of single-use packaging (plastic or glass), which can be easily avoided by providing different options and making a little bit of extra effort. The result on the environment is surely significant for the aspect of ecosystem integrity and is an important step to accompany any sustainable initiative. In terms of impacts, saving materials can be as effective as points 5-6 or even more so, depending on the amount of packaging and plastic used per person.

In Tables 6 and 7, we provide some examples of actions within Sustainable Diet Initiatives carried out by universities or major catering firms. Most of the 8 SU-EATABLE principles are met in many of the actions presented.

Table 6 – Examples of university initiatives (in italics) on health/sustainable diets carried out at their canteens (bold)

Kings College, London UK (SRA member) <i>Sustainability Project Funds</i> Target: environmental sustainability as a whole Actions: Reducing Single-Use Disposables by outlets selling reusable Keep Cups, using disposable cutlery, bringing your own, save 20p per hot drink purchased in your reusable cup afterwards. Organic Food, Fairtrade Certification, Buy Local; Reducing Meat Consumption, part time-carnivore initiative (https://www.parttimecarnivore.org/), vegan menu; Reducing Food Waste, food waste calculator for personal waste quantity and impact evaluation.
University of Portsmouth UK <i>“Kale Yeah!” initiative</i> Target: Low C emission diets and Healthy Eating Actions: Reducing meat, fish and dairy consumption. Encouraging vegetarian or vegan options by accumulating scores on an electronic loyalty card to get a free meal, fun and punchy promotion campaigns.



Harvard University USA <i>Harvard's Sustainable and Healthful Food Standards</i> Target: Sustainable food choices, waste reduction, animal welfare, workers welfare, promoting education Actions: Promoting diets with less meat and more vegetables, purchasing only sustainable seafood, reducing food waste, reducing single-use packaging, reducing antibiotic/chemical use in food, promoting the welfare of workers and animals along the food chain
University of Leiden, NL <i>Pure concept</i> Target: Environment, Health Actions: Affordable price tag for Pure products that have additional value such as being less harmful to the environment (vegetarian food), healthier (less salt, sugar, fat), Bio-based material for canteen meals and take-away, "Green Snack Factory" vending machine with only healthy and sustainable food, and the water jar with tap water with fragrant herbs and refreshing fruit added for flavour, plus 40% of the vitamins

Table 7 – Examples of initiatives (in italics) on health/sustainable diets carried out by the catering sector (bold) at different canteen sites (public/private)

Vacherin Caterer UK <i>"I'm Perfect", "Waste no Food", "A Greener Choice"</i> Targets: Reduce food waste, Lower C emissions, Reduce avoidable waste Actions: New recipes and food offers based on sourced products generally wasted, like 'wonky' fruit and vegetables, or food destined for waste like leftover porridge and pulp from smoothies. Reduce waste by setting targets and measuring against them. Strict control to ensure that any unavoidable waste reaches the most sustainable end-point. Reusable and recyclable alternatives to single-use disposables. Low-carbon menu offer with less meat. Results: Profit maximisation to head office; creativity exercise for chefs; successful link of chef demand to farmer offer of wonky fruit and veg; increasing sales on repurposed food
Artizian Caterer UK <i>"Plant Power 2018", "Mood Food"</i> Targets: Sustainable and Healthy Eating Actions: Reducing meat consumption and increasing fruit and vegetable intake by placing veggie options at the top of the menu, changing the layout look and location of certain products, increasing plant-based seasonally-led options of hot food and grab and go menu, Meat Free Mondays, Initiatives supported by marketing, POS systems, keeping choices exciting and constant array of new menus, educating and encouraging people to use what they've learnt at work to cook at home. Results: Plant power dishes sold increased to 46% of the dishes from a start point of just 21%



CH&CO Ireland/UK Planetary program

Target: Environment, Health, Animal welfare

Actions: More vegetarian options, one-fourth of options are plant-led; meat free Mondays; having vegetarian chefs train canteen chefs; cage free eggs; serving sustainably caught fish; healthy alternatives, sugar and salt reduction; grab&go food paired with my fitness pal app; car fleet CO2 emissions reduced by 10G / 9% - thanks to electric and hybrid cars; all staff completed online planet training; smart meters have been installed in head offices; switched to LED lighting; a comprehensive range of environmentally-friendly cleaning products; influencing clients on waste streams; daily specials and nose-to-tail cookery; reviewed portion sizes; takeaway boxes for customers; redistribution of food to client-selected charity; recycling coffee grounds and cooking oil; preserving food items where possible; site-made stocks and flavoured oils; chef training to minimise waste; food waste monitoring; 99% of water sales are in a can, glass or PET (recyclable plastic); glass jars with clip-on lids; paper bags; paper straws; wine in a can, bottle or PET (recyclable plastic).

Results: not reported



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Appendix I - 8 key principles: explanation boxes

The 8 key principles of the SU-EATABLE DIET – THE WHY

The following boxes report the scientific evidence in support of the 8 principles on which the SU-EATABLE LIFE diet is based. Environmental impact is specifically addressed with reference to the two main indicators of the SU-EATABLE Life Project: CO₂ emissions equivalents, and water quality and quantity. The amount of CO₂ emitted per kg or litre of food product is referred to as Carbon Footprint (question box n°2), or is reported as CO₂ equivalents (question box n°4). The amount of water used per kg or litre of food product is indicated as Water Footprint – WF (question box n°3).

1. Champion plant-based food in your diet

Environment: Plant-based food emits on average significantly less CO₂ equivalents (0.2 to 2.7 kg CO₂-eq/kg) compared to animal-based food (3.5 to 28 kg CO₂-eq/kg), with the exception of milk and eggs, which are in the range of plant-based food (Clune et al. 2017, Poore and Nemecek 2018, Willet et al. 2019). Large savings in WF with dietary choices can be achieved depending on the weekly menu. The WF of vegetables, grains and fruit is about 25 times lower when compared to beef; as a protein source, legumes have a WF from 2 to 4 times higher than the beef WF and similar to the WF of chicken and eggs (Mekonnen, M.M. and Hoekstra, A.Y., 2010). The optimal daily diet [-composition-] for health and the environment which could help the planet to stay within its sustainable resource use should consist, by volume, of approximately half a plate of vegetables and fruits, and a significant source of protein should still be provided by legumes, nuts and cereals (Willet et al. 2019). To date, in industrialized countries, people tend to consume too much sugar, oils and fats, red meat as well as milk and cheese combined, while not enough fruit and vegetables (Vanham et al 2018, Willet et al. 2019).

Health: According to the World Health Organization, a healthy diet includes: fruit, vegetables, legumes, nuts and whole grains, with at least 400 g (i.e. five portions) of fruit and vegetables per day (WHO, 2018). Emphasizing nutrient-dense, plant-based foods such as fruit, vegetables, legumes and nuts, is associated with significantly better health outcomes (Tuso et al. 2013; Satija et al., 2016; BDA, 2017; Satija et al., 2017; Toumpanakis et al. 2018).

2. Enjoy meat in moderate amounts

Environment: Meat production significantly affects climate change, as the “impacts of the lowest-impact animal products typically exceed those of vegetable substitutes” (Poore and Nemecek 2018). At the highest end of CO₂-eq emissions is ruminant meat, i.e. beef and lamb, which can be almost 10 times higher than poultry, eggs, rabbit and most fish, and 4 to 5 times higher than pork (Clune et al. 2017, Poore and Nemecek 2018). In terms of pressure on freshwater resources, meals that include beef have 3 times the WF as those that include other meat protein sources (pork, chicken) or legumes, and twice the WF of eggs (Mekonnen, M.M. and Hoekstra, A.Y., 2010). The big WF impact of beef is mainly related to the production of animal feed over the lifetime of the animal (Hoekstra 2015). A WF reduction of up to 35% (average per capita per day) could be achieved by adopting the common guidelines for healthy diets (Vanham et al. 2018). A good compromise for protein intake is represented by fish, both in terms of CO₂ equivalent emissions and water saving (Mekonnen, M.M. and Hoekstra, A.Y., 2010, Clune et al. 2017). A full shift to a pescatarian (or vegetarian) diet could reduce the WF per capita within the range 33-55% (Vanham et al. 2018).

Health: Following reclassification in 2015 by the World Health Organization’s International Agency for Research on Cancer, processed meats (such as hot dogs, ham, sausages, corned beef, canned meat, and meat-based sauces) eaten daily may increase the risk of colorectal cancer and stomach cancer. A minimum intake of these types of food that guarantees complete safety cannot be defined. Similarly, also red meat consumption (i.e. all muscle meat, such as beef, veal, pork, and lamb) is associated with an increased risk of colorectal, pancreatic, and prostate cancers (IARC, 2015), cardiovascular disease (Abete et al. 2014), and risk of stroke and type 2 diabetes (Chen et al., 2013, Feskens et al., 2012). Such risks can be significantly diminished by substituting other protein sources for red meat (Pan et al., 2012).

According to the EAT-Lancet commission (Willett et al. 2019): “Because intake of red meat is not essential and appears to be linearly related to total mortality and risks of other health outcomes in populations that have consumed it for many years, optimal intake might be 0 g/day, especially if replaced by plant sources of protein. Because data on risk of low intakes of red meat are imprecise, we have concluded that an intake of 0 g/day to about 28 g/day of red meat is desirable and have used a midpoint of 14 g/day for the reference diet. Since consumption of poultry has been associated with better health outcomes than red meat, we have concluded that the optimum consumption of poultry is 0 g/day to about 58 g/day and have used a midpoint of 29 g/day for the reference diet.”

3. Enjoy dairy products in moderation

Environment: Plant-based milk, except for rice milk, has a lower Carbon Footprint (CF) than dairy milk (Clune et al. 2017). The difference is even more significant for the Water Footprint (WF of soy milk is 296.9 L/kg vs cow's milk 1050 L/kg, Ercin et al. 2011). Yogurt has a value similar to milk, while cheese - due to the high concentration of milk used for its production - can have a CF and WF up to 6 times higher than the original milk. When compared with other protein sources, cheese has twice the WF of soy-based proteins, and both the CF and WF of cheese can be higher than those of many meat sources, including chicken, pork and fish (Clune et al. 2017, Mekonnen, M.M. and Hoekstra, A.Y., 2010).

Health: High intake of dairy products, at least three servings per day, has been widely promoted in Western countries for bone health and fracture prevention, primarily because of their high calcium content. However, the optimum calcium intake remains uncertain (Willett et al., 2019), while there is evidence that the adverse effect of protein of animal origin might outweigh the positive effect of calcium intake on the calcium balance (FAO and WHO, 2002). Animal-based dairy fat like butter conveys a higher risk of cardiovascular disease than unsaturated plant oils, which therefore should be preferred whenever possible for cooking and food preparation (EAT-Lancet Commission, Willett et al., 2019).

4. Avoid too much food

Environment: Every kilogram of food carries the embedded environmental impact generated during its production, processing, transportation, and packaging (Willett et al. 2019, Ercin et al. 2011, Clune et al. 2017, Vanham and Bigoglio, 2013). To reduce pressure on the environment, we need to not only balance the composition of our diet, but also to reduce the portions of food we consume. Every time we throw food into the waste bin, we are basically wasting resources and generating unnecessary environmental pressures. The wasted environmental value is higher for meat and animal-based food (Clune et al. 2017, Willett et al. 2019). A dietary shift combined with reducing our food waste might help us to reach important sustainability targets in terms of climate change and water use (Willett et al. 2019). In Spain, for example, the WF of food waste accounts for 2095 hm³, equivalent to 131 litres per person per day (Blas et al. 2018).

Health: Energy intake (calories) should be in balance with energy expenditure (WHO, 2018). Excess energy intake (i.e. energy-dense unhealthy dietary patterns) has been identified as the most significant dietary factor in relation to weight gain and the development of obesity (Roberts et al., 2002).

5. Celebrate variety

Environment: Agrobiodiversity is an important strategy for facing the challenges of climate change, as it offers more flexible climate adaptation strategies, thus supporting food security and resource saving (Fischlin et al. 2007). Widening food choices allows for more creativity in our diet, making it easier to plan more sustainable diets. Coupling variety with highlighting seasonal and local products could help to boost local economies and save resources. Seasonality is also important. The CF of vegetables and fruit grown in heated greenhouses can be increased four-fold (Clune et al. 2017). Refrigerated transport of plant-based food and long-term refrigerated storage of “out of season” vegetables and fruit in warehouses can be a significant source of CO₂ equivalents (Hospido et al. 2009, James & Jasmes 2010, Wakeland et al. 2012). Generally, only when local markets are affected by seasonal constraints which strongly limit food variety (cold climates), long-distance supply chains may be more advantageous compared with greenhouse food production or with long-term food storage throughout the seasons (Wakeland et al. 2012).

Health: Dietary diversity has long been recognized as a key element of high-quality diets. Increasing the variety of foods across and within food groups is recommended in most dietary guidelines internationally, as a means to ensure adequate intake of essential nutrients and to promote good health (Ruel, 2003). Simplification of human diets associated with increased accessibility of inexpensive commodities and erosion of agrobiodiversity lead to nutrient deficiencies and excess energy consumption (Johns & Eyzaguirre, 2006). Fruits, minor vegetables, and leaves used as condiments, spices, or sauce ingredients can be grown in small quantities and add variety and essential nutrients to diets otherwise dominated by staple foods. Agrobiodiversity is an underexplored avenue for giving both food producers and consumers access to greater dietary diversity (Johns et al. 2006).

6. Fresh is best (for you and for the environment)

Environment: Food processing can add further pressure on the environment. The more complex the processing, the higher amount of energy and materials required, which leads to higher CO₂-eq emissions and resource consumption per kilogram of product (Notarnicola et al. 2017). For example, the CF of fresh peas can increase by 60% to more than 100% when purchased canned, jarred or frozen (Clune et al. 2017, Carbon Footprint Frozen Food Report 2012). The global warming potential (GWP) of processed tomatoes (chopped, peeled, puréed) can be twice the amount of fresh tomatoes due to the energy use for processing and packaging (Del Borghi et al. 2014).

Health: The increasing proportion of ultra-processed foods in diets has been identified as a driver of excess energy intake (Monteiro, 2010). Also, the evidence thus far shows that displacement of minimally processed foods and freshly prepared dishes and meals by ultra-processed products is associated with unhealthy dietary nutrient profiles. (Monteiro, Cannon et al. 2018).

7. Drink tap water rather than bottled water whenever possible and safe

Environment: When safe tap water is available, the use of water in plastic bottles places an unnecessary burden on the planet. An analysis of 6 different Italian water providers showed that consumers can save on a yearly basis 163.50 kg of CO₂-eq per capita by drinking 2 litres of tap water rather than PET-bottle water (Botto et al. 2011). In terms of water savings, bottling 1.5L of water requires around 1.9 extra litres of water for bottling operations, processes, packaging and transport (Nicolucci et al. 2011), compared with 0.5 litres of extra water if using tap water responsibly from efficient municipal systems. Bottled water in plastic material poses a further environmental burden, as plastic production relies heavily on oil (Foolmaun & Ramjeawon 2008), and in the EU the amount of plastic recycled in many countries is still less than 50% of the production (Plastic Europe 2017).

Health: Good hydration is vital for good health and well-being. Water safety and quality are fundamental to human development and well-being, and 71% of the global population uses a safely managed drinking-water service (WHO, 2017). The reason for the increase of bottled water consumption is not straightforward; one of the most common factors seems to be consumer dissatisfaction with tap water organoleptics, i.e. the water characteristics that affect the senses of taste, odour and sight (Miguel, 2006). However, the quality of bottled water is not necessarily superior when compared with municipal tap water, as shown by Rosa et al. (2011) in a study conducted on 37 bottled waters available on the market and 15 municipal water pipelines.



8.Reduce single-use!

Environment: Food packaging represents an important fraction (25%) of the plastic materials that leak into the environment. The estimated annual input of mismanaged plastic waste to the ocean is about 13 million metric tons (Jambeck 2015). This plastic remains in the ocean for centuries, causing harm to natural systems. Shopping for food in hypermarkets has shown to have 3 times the CF per food basket than shopping in local food markets, due to the higher amount of packaging in hypermarket food (Sanyé et al. 2012). Retailers could save a great deal of CO₂ emissions by helping to reduce food packaging, while users could significantly reduce the environmental footprint of their food basket by recycling or bringing their own shopping bags (Sanyé et al. 2012).

Appendix II - SU-EATABLE LIFE Survey

The SU-EATABLE LIFE survey was implemented using Formstack and was offered in English and Italian. It consists of two sections (questions on sustainable diets – reported in full below – and demographics), preceded by an initial page explaining to respondents that their answers would be used for research purposes only, in aggregated form and in accordance with the European Privacy and Data Protection laws. All the respondents had to express their informed consent to participate prior to being allowed to proceed to the next sections of the survey.

The survey was piloted internally and then refined, and remained available online from February to March 2019. In addition to its promotion on the SU-EATABLE LIFE website and newsletter, it was circulated via the project beneficiaries' digital channels (websites, social media pages, newsletters, intranet).

Furthermore, various public and private organizations were contacted and accepted to help in publicizing the survey to their staff, members, readers and affiliates via different channels (among others, providing a link to the survey from their website, through their intranet, using their mailing lists and newsletters, and publishing news and posts on their social media pages). Two Italian e-journals published dedicated articles describing the SU-EATABLE project and inviting the readers to take the survey.

In total, 812 people responded to the survey. Ten responses were excluded since they were incomplete, resulting in a sample comprising 802 respondents, 70% of whom were women and 30% were men, aged 52 on average. Out of all respondents, the majority were omnivores (63%), 23% of were flexitarian and the remaining 14% were either vegetarian, vegan or pescatarian.

SU-EATABLE LIFE survey. List of questions.

Q1: Which of these concepts do you associate with the idea of '**sustainable diets**'?

Please select up to 5 options

- ☐ Inexpensive food
- ☐ Healthy food
- ☐ Good for the environment
- ☐ Protections for farmers and agricultural workers
- ☐ Low meat consumption
- ☐ Vegetarian
- ☐ Vegan
- ☐ Low emissions of greenhouse gases
- ☐ Low-intensity water usage
- ☐ Preserves and encourages biodiversity
- ☐ Certification based on location (DOC, DOP, etc.)
- ☐ Certification based on welfare (Red Tractor, Free-Range, etc.)



- ☐ Certification based on farming practices (Organic, Bio, LEAF, etc.)
- ☐ Minimal or no chemical use
- ☐ Local food
- ☐ Traditional foods
- ☐ Wild fish
- ☐ Free-range meat and dairy
- ☐ Biodegradable packaging
- ☐ Recyclable packaging
- ☐ Farmers markets
- ☐ Sustainable retailers or health food stores

In the questions below, “retailers” refers to local markets, supermarkets, local shops, etc. “Food service providers” refers to canteens, take-aways, restaurants etc.

Q2: When you buy food at retailers and/or food service providers, do you feel confident that you are able to recognise goods that are produced with **low greenhouse gas emissions**?

- ☐ Always
- ☐ Most of the time
- ☐ Occasionally
- ☐ Hardly ever
- ☐ Never

Q3: When you buy food at retailers and/or food service providers, do you feel confident that you are able to recognise goods that are produced with **low water consumption**?

- ☐ Always
- ☐ Most of the time
- ☐ Occasionally
- ☐ Hardly ever
- ☐ Never

Q4: When you buy food at retailers and/or food service providers, do you feel confident that you are able to recognise goods that are produced encouraging **biodiversity and/or environmental protections**?

- ☐ Always
- ☐ Most of the time
- ☐ Occasionally
- ☐ Hardly ever
- ☐ Never

Q5: When you buy food at retailers and/or food service providers, are you able to identify if goods are produced using **local or heritage ingredients**?

- ☐ Always
- ☐ Most of the time

- ☐ Occasionally
- ☐ Hardly ever
- ☐ Never

Q6: When you buy food at retailers and/or food service providers, do you know which products are **in season**?

- ☐ Always
- ☐ Most of the time
- ☐ Occasionally
- ☐ Hardly ever
- ☐ Never

Q7: What are your priorities when buying food?

Please select up to 3 options

- ☐ Cost
- ☐ Traceability
- ☐ Ease in cooking/preparation
- ☐ Health impact
- ☐ Environmental Impact
- ☐ Quality
- ☐ Seasonality

Q8: Are you informed on climate change issues and risks?

- ☐ Very well informed
- ☐ Quite well informed
- ☐ I understand a bit, but not a lot
- ☐ I don't know or understand much about it
- ☐ I don't know anything about it beyond some simple key issues
- ☐ I know nothing about it

Q9: Would you consider yourself informed on sustainable food production and agriculture?

- ☐ Very well informed
- ☐ Quite well informed
- ☐ I understand a bit, but not a lot
- ☐ I don't know or understand much about it
- ☐ I don't know anything about it beyond some simple facts
- ☐ I know nothing about it

Q10: Do you think that through food choice citizens might reduce their personal impact on the



environment?

- ☐ Yes
- ☐ No
- ☐ I don't know

Q11: Would you be willing to pay more for food with a low impact on the environment?

- ☐ Yes
- ☐ No
- ☐ Only if cost increase is below 10%
- ☐ Only if cost increase is below 30%

Q12: Are you interested in learning more about how your food choices influence the environment?

- ☐ No
- ☐ Yes

Q13: How would you like to become more informed?

Please select up to 3 options

- ☐ By information displayed in a shop/canteen/restaurant/other
 - ☐ By joining a group of people with similar interests
 - ☐ Through a dedicated website
 - ☐ Through social media
 - ☐ Through an app
 - ☐ Through a newsletter/email service
 - ☐ Through a video
 - ☐ Other (please specify):
-

Appendix III - SU-EATABLE LIFE Survey results

Table 2 - Survey results – Results of the questionnaires carried out in UK and IT with 13+4 questions, expressed as a percentage of the total scores from UK plus Italy (802)

Q1 Which of these concepts do you associate with the idea of ‘sustainable diets’?			
Inexpensive food	0.7%	Certification based on location	1.9%
Healthy food	5.6%	Certification based on welfare	3.3%
Good for the environment	10.8%	Certification based on farming practices	3.9%
Protections for farmers and agricultural workers	5.0%	Minimal or no chemical use	6.1%
Low meat consumption	6.0%	Local food	5.7%
Vegetarian	1.9%	Traditional foods	1.2%
Vegan	1.4%	Wild fish	0.9%
Low GHG emissions	9.6%	Free-range meat and dairy	3.8%
Low-intensity water usage	7.1%	Biodegradable packaging	6.2%
Preserves and encourages biodiversity	8.2%	Recyclable packaging	5.4%
		Farmers markets	2.1%
		Sustainable retailers or health food stores	3.1%
Q2 When you buy food at retailers and/or food service providers, do you feel confident that you are able to recognise goods that are produced with low greenhouse gas emissions?		Q3 When you buy food at retailers and/or food service providers, do you feel confident that you are able to recognise goods that are produced with low water consumption?	
Always	0.6%	Always	0.6%
Most of the time	8.9%	Most of the time	4.4%
Occasionally	23.2%	Occasionally	14.5%
Hardly ever	42.1%	Hardly ever	43.1%
Never	25.2%	Never	37.4%
Q4 When you buy food at retailers and/or food service providers, do you feel confident that you are able to recognise goods that are produced encouraging biodiversity and/or environmental protections?		Q5 When you buy food at retailers and/or food service providers, do you feel confident that you are able to recognise goods that are produced using local or heritage ingredients?	

Always	1.4%	Always	2.6%
Most of the time	9.1%	Most of the time	33.4%
Occasionally	35.9%	Occasionally	47.6%
Hardly ever	35.7%	Hardly ever	12.6%
Never	18.0%	Never	3.7%
Q6 When you buy food at retailers and/or food service providers, do you know which products are in season?		Q7 What are your priorities when buying food?	
Always	16.8%	Cost	15.1%
Most of the time	48.1%	Traceability	6.0%
Occasionally	26.1%	Ease in cooking/preparation	6.4%
Hardly ever	6.7%	Health impact	22.0%
Never	2.2%	Environmental Impact	13.4%
		Quality	22.6%
		Seasonality	14.4%
Q8 Are you well informed on climate change issues and risks?		Q9 Would you consider yourself informed on sustainable food production and agriculture?	
Very well informed	28.2%	Very well informed	13.8%
Quite well informed	48.5%	Quite well informed	38.4%
I understand a bit, but not a lot	21.7%	I understand a bit, but not a lot	35.5%
I don't know much about it	0.4%	I don't know much about it	5.1%
I don't know anything about it beyond some simple key issues	1.1%	I don't know anything about it beyond some simple key issues	6.4%
I know nothing about it	0.1%	I know nothing about it	0.7%
Q10 Do you think that through their food choice citizens might reduce their personal impact on the environment?		Q11 Would you be willing to pay more for food with a low impact on the environment?	
Yes	94.1%	Yes	54.5%
No	1.2%	No	5.1%
I don't know	4.4%	Only if cost increase is below 10%	22.4%
Empty	0.2%	Only if cost increase is below 30%	18.0%
Q12 Are you interested in learning more about how your food choices influence the environment?			
Yes	92.1%		
No	7.9%		



Q13 How would you like to become more informed?					
By information displayed in a shop/canteen/restaurant/other			22.3%		
By joining a group of people with similar interest			6.0%		
Through a dedicated website			13.0%		
Through social media			12.4%		
Through an app			10.4%		
Through a newsletter/email service			7.3%		
Through a video			6.0%		
Other (please specify)			22.7%		
Q14 Age distribution by age group			Q15 Gender		
(1-10)	0.0%	(51-60)	11.8%	male	27.9%
(11-20)	1.6%	(61-70)	6.0%	female	70.9%
(21-30)	38.5%	(71-80)	1.2%	prefer not to answer	0.9%
(31-40)	26.4%	(81-90)	0.0%	non-binary	0.2%
(41-50)	14.3%	(91-100)	0.0%		
Q16 Which of the following best describes your current position?			Q17 What are your dietary habits?		
Employed (full-time)	54.9%	Omnivore (you eat everything)	63.1%		
Employed (part-time)	13.3%	Flexitarian	23.4%		
Student	17.3%	Vegetarian	8.5%		
Retired	4.5%	Vegan	2.2%		
Unemployed	2.7%	Pescatarian	1.5%		
Other	7.2%	Other	1.2%		

Appendix IV - Question boxes

Question box 1 – What does “Climate Change” mean?

Definition: The term Climate Change indicates a long-term variation or trend of change in climate properties, like air temperature, humidity, atmospheric pressure, precipitation, and winds, identified by statistical analysis over extended periods (decades or longer) (Cubasch et al. 2013). These properties are also used to commonly define the weather. The climate of a region or country represents the average weather over a long period of time, generally at least 30 years (World Meteorological Organization). One of the most important properties used to describe the on-going trend of climate change is the global average atmospheric temperature.

A commonly used definition of Climate Change, related to its actual causes, is given by the Framework Convention on Climate Change (UNFCCC, 1992) in its Article 1: “a change of climate which is attributed directly or indirectly to human activity that alters the composition of the global atmosphere and which is in addition to natural climate variability observed over comparable time periods.”

Current trend: The global average atmospheric temperature has been constantly increasing since record-keeping began in 1850. This phenomenon is known as Global Warming. At present it is approximately 1°C higher than it was in the late 19th century (IPCC, 2018). The most recent climatic simulations indicate a threshold of +2° compared to pre-industrial times as the limit above which there will be a high risk of dangerous and catastrophic environmental changes on the local and global scale (IPCC, 2018).

Causes: Although climate change can be induced by natural causes, like changes in solar cycles, volcanic eruptions, and so on, there is a strong general consensus among the scientific community that the main cause of the current global warming is anthropogenic activity that is introducing into the atmosphere high amounts of long-lived greenhouse gases – GHG (CO₂, CH₄, N₂O, CFC, O₃, etc.) as the consequence of human activities which make use of fossil fuel energy sources, deforestation, agricultural management and fertilizer use, livestock farming, and use of fluorinated gases (Myhre, G., et al. 2013). Additionally, deforestation and land degradation have strongly reduced the natural sink of atmospheric CO₂, which contributes to maintaining the atmospheric balance of this important greenhouse gas. If the human population continues to maintain its present habits, the projections of world GHG emissions indicate that by the year 2100 the planet will pass the critical threshold of 2°C in the average atmospheric temperature increase, which will probably settle at around 3-4°C, with catastrophic consequences for the planet and its inhabitants (IPCC, 2018).

Question box 2 – What does “Carbon Footprint” mean?

Carbon footprint is a term that describes the amount of greenhouse gases (GHGs) that have been emitted by all the processes involved in the production of a good or a service (PAS 2050, ISO 14064-1:2018, GHG protocol). For goods, the Carbon Footprint of a product is the amount of CO₂ equivalents produced (see box 3) per unit of production. This unit could be a weight measurement, like a kilo or litre, or a unit of production such as a wine bottle or a pot of yogurt.

When calculating the Carbon Footprint of a good, all the processes from 'cradle to grave' are considered. This includes, but is not limited to, the raw materials, production, processing, packaging, distribution, and all the way to the end life of the waste products used to create the item.

Looking at this journey from cradle to grave is called 'Life Cycle Analysis' (ISO 14040:2006). This allows us to calculate the total GHG emissions from a product's creation, including the energy source used for transportation, the impact from all materials used (such as fertilizers or packaging), livestock or soil, and waste management.

Question box 3 – What does “Water Footprint” mean?

The Water Footprint (Hoekstra and Chapagain 2008; Hoestra 2015) is an indicator of water use in relation to consumer goods and is similar to the concept of ecological and carbon footprints. The water footprint of a product is the volume of fresh water used to produce the product along the production chain.

The WF has three components:

- The blue water footprint is the volume of fresh water that has evaporated from the global blue water resources (surface and groundwater).
- The green water footprint is the volume of water evaporated from the global green water resources (rainwater stored in the soil).
- The grey water footprint is the volume of water that would be required to dilute pollutants generated during the production process so as to maintain the quality of the ambient water within local water quality standards.

As the three elements of the water footprint depend on local climate, hydrology, crop types, and other environmental pressures present in the same water basin, the water footprint is a geographically explicit indicator, so it can significantly vary for the same food item depending on the place of production.

Question box 4 – What does “CO₂ equivalents” mean?

All greenhouse gases (GHGs) have the capacity to absorb and emit radiant energy in the thermal infrared wavelength range. This means that they play a key role in trapping thermal energy emitted by the Earth within its atmospheric boundaries. This results in an increase of the average atmospheric temperature, i.e. “global warming”.

As GHGs all act through the same mechanism, we can use a common scale called the “CO₂ equivalent emissions” (Shine, 2009). This can be calculated using the Global Warming Potential (GWP) of each gas, which is the known relative radiative effect of the gas against the effect of carbon dioxide (CO₂).

For example, on a mass basis, the GWPs of the gases CH₄ and N₂O are 34 times and 298 times higher than the GWP of CO₂, respectively (100 years projection, IPCC 2013).

This means that 1 g of CH₄ is equivalent in terms of greenhouse effect to 34 g of CO₂, and 1 g of N₂O is equivalent to 298 g of CO₂. In this way we can sum up the three gases using their equivalent values in terms of CO₂, which we call CO₂ equivalents (CO₂e):

$$\begin{aligned}
 &1\text{g CH}_4 + 1\text{g N}_2\text{O} + 20\text{g CO}_2 = \\
 &1\text{g CH}_4 \times 34 + 1\text{g N}_2\text{O} \times 298 + 20\text{g CO}_2 \times 1 = \\
 &24\text{g CO}_2\text{e} + 310\text{g CO}_2\text{e} + 20\text{g CO}_2 = \\
 &323\text{g CO}_2\text{e}
 \end{aligned}$$

Note: “The GWP for a time horizon of 100 years was later adopted as a metric to implement the multi-gas approach embedded in the United Nations Framework Convention on Climate Change (UNFCCC) and made operational in the 1997 Kyoto Protocol.” (IPCC, 2013)